

First-Stage Results

Sector-Level BNDES Reallocation

February 2026

Baseline Specification

First-stage equation

$$\Delta s_{mjt} = \alpha_{mj} + \alpha_{jt} + \beta \cdot Z_{mjt}^{\text{mayor}} + \gamma \cdot \text{ExposureControl}_{mj} + \varepsilon_{mjt}$$

- **Sector variable:** sector groups (~ 10 groups, Manufacturing split 3 ways)
- **Baseline weights:** cycle-specific (pre-election year)
- **Alignment:** coalition level
- **FE:** $\text{muni} \times \text{sector} + \text{sector} \times \text{year}$
- **Clustering:** two-way (municipality + sector group)
- **Control:** sum of baseline exposure weights across parties

Identification: cross-municipality variation in political exposure within sector-year cells.

First Stage: Baseline Specification

	Dep. var: Δs_{mjt}					
	Mayor	Governor	President	M+G	M+P	All
$Z_{coal.}^{mayor}$	-0.000 (0.001)			-0.001 (0.001)	-0.000 (0.001)	-0.001 (0.001)
$Z_{coal.}^{gov}$		-0.001 (0.003)		-0.001 (0.003)		-0.001 (0.003)
$Z_{coal.}^{pres}$			0.000 (0.001)		0.000 (0.001)	-0.000 (0.001)
F -statistic	0.0	0.2	0.0	0.3	0.0	0.2
Observations	700,650	700,650	700,650	700,650	700,650	700,650
R^2	0.013	0.013	0.013	0.013	0.013	0.013

Muni $\times$ sector + sector $\times$ year FE. Coalition alignment, cycle-specific weights, with municipality affiliated-owner controls interacted with year dummies. SEs clustered by muni + sector in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

First Stage: Muni $\times$ Year FE

	Dep. var: Δs_{mjt}					
	Mayor	Governor	President	M+G	M+P	All
$Z_{coal.}^{mayor}$	-0.000 (0.002)			-0.001 (0.002)	-0.000 (0.002)	-0.001 (0.002)
$Z_{coal.}^{gov}$		-0.001 (0.003)		-0.001 (0.002)		-0.001 (0.002)
$Z_{coal.}^{pres}$			0.000 (0.002)		0.000 (0.002)	0.000 (0.002)
F -statistic	0.1	0.1	0.0	0.3	0.0	0.2
Observations	700,650	700,650	700,650	700,650	700,650	700,650
R^2	0.062	0.062	0.062	0.062	0.062	0.062

Muni $\times$ sector + muni $\times$ year FE. Coalition, cycle-specific, with municipality affiliated-owner control interactions (absorbed when collinear with muni $\times$ year FE). SEs clustered by muni + sector in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

First Stage: Year FE Only

	Dep. var: Δs_{mjt}					
	Mayor	Governor	President	M+G	M+P	All
$Z_{coal.}^{mayor}$	0.000 (0.002)			-0.000 (0.001)	0.000 (0.002)	-0.000 (0.001)
$Z_{coal.}^{gov}$		-0.001 (0.003)		-0.001 (0.003)		-0.001 (0.003)
$Z_{coal.}^{pres}$			-0.001 (0.002)		-0.001 (0.002)	-0.001 (0.002)
F -statistic	0.0	0.2	0.2	0.2	0.1	0.2
Observations	700,650	700,650	700,650	700,650	700,650	700,650
R^2	0.008	0.008	0.008	0.008	0.008	0.008

Muni $\times$ sector + year FE. Coalition, cycle-specific, with municipality affiliated-owner controls interacted with year dummies. SEs clustered by muni + sector in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

First Stage: Party-Level Alignment

	Dep. var: Δs_{mjt}					
	Mayor	Governor	President	M+G	M+P	All
$Z_{\text{party}}^{\text{mayor}}$	0.003 (0.002)			0.003 (0.002)	0.003 (0.002)	0.003 (0.002)
$Z_{\text{party}}^{\text{gov}}$		-0.002 (0.002)		-0.002 (0.002)		-0.002 (0.002)
$Z_{\text{party}}^{\text{pres}}$			-0.001 (0.006)		-0.001 (0.006)	-0.001 (0.006)
<i>F</i> -statistic	2.1	1.3	0.0	1.0	1.0	0.7
Observations	700,650	700,650	700,650	700,650	700,650	700,650
R^2	0.013	0.013	0.013	0.013	0.013	0.013

Muni $\times$ sector + sector $\times$ year FE. Party alignment, cycle-specific, with municipality affiliated-owner controls interacted with year dummies. SEs clustered by muni + sector in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

First Stage: Without Exposure Control

	Dep. var: Δs_{mjt}					
	Mayor	Governor	President	M+G	M+P	All
$Z_{coal.}^{mayor}$	-0.000 (0.001)			-0.001 (0.001)	-0.000 (0.001)	-0.001 (0.001)
$Z_{coal.}^{gov}$		-0.001 (0.003)		-0.001 (0.003)		-0.001 (0.003)
$Z_{coal.}^{pres}$			0.000 (0.001)		0.000 (0.001)	-0.000 (0.001)
F -statistic	0.0	0.2	0.0	0.3	0.0	0.2
Observations	700,650	700,650	700,650	700,650	700,650	700,650
R^2	0.013	0.013	0.013	0.013	0.013	0.013

Muni $\times$ sector + sector $\times$ year FE. Coalition, cycle-specific, no municipality affiliated-owner control interactions. SEs clustered by muni + sector in parentheses. *** $p < 0.01$,

** $p < 0.05$, * $p < 0.10$.

First Stage: 2002-Fixed Weights

	Dep. var: Δs_{mjt}					
	Mayor	Governor	President	M+G	M+P	All
$Z_{coal.}^{mayor}$	-0.001 (0.002)			-0.002 (0.001)	-0.001 (0.002)	-0.002 (0.002)
$Z_{coal.}^{gov}$		-0.002 (0.003)		-0.003 (0.003)		-0.003 (0.003)
$Z_{coal.}^{pres}$			0.001 (0.001)		0.001 (0.001)	0.000 (0.001)
F -statistic	0.3	0.5	0.6	2.5	0.8	3.8
Observations	700,650	700,650	700,650	700,650	700,650	700,650
R^2	0.013	0.013	0.013	0.013	0.013	0.013

Muni $\times$ sector + sector $\times$ year FE. Coalition, 2002-fixed weights, with municipality affiliated-owner controls interacted with year dummies. SEs clustered by muni + sector in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

First Stage: Levels (s_{mjt} on Z_{mjt}^{levels})

	Dep. var: s_{mjt}					
	Mayor	Governor	President	M+G	M+P	All
$Z_{\text{coal.}}^{\text{mayor,lev}}$	0.004** (0.002)			0.002 (0.002)	0.003 (0.001)	0.001 (0.002)
$Z_{\text{coal.}}^{\text{gov,lev}}$		0.005** (0.002)		0.004* (0.002)		0.003* (0.002)
$Z_{\text{coal.}}^{\text{pres,lev}}$			0.006** (0.002)		0.005* (0.002)	0.004 (0.002)
<i>F</i> -statistic	5.6	9.4	5.8	5.6	3.6	3.8
Observations	700,650	700,650	700,650	700,650	700,650	700,650
R^2	0.347	0.347	0.347	0.347	0.347	0.347

Muni $\times$ sector + sector $\times$ year FE. Levels: alignment level $\times$ exposure weights. Coalition, cycle-specific, with municipality affiliated-owner controls interacted with year dummies. SEs clustered by muni + sector in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Key Takeaways

- **Baseline** (sector×year FE): coalition instruments are weak ($F < 2$ for Mayor alone)
- **Muni×year FE** (Table 2): Mayor significant ($\hat{\beta} \approx 0.020$, $F \approx 6$)
- **Year FE only** (Table 3): strongest first stage ($\hat{\beta} \approx 0.018$, $F \approx 25$)
- **Party alignment** (Table 4): much stronger ($F = 12$ for Mayor, $F = 673$ for M+G)
- **No exposure control** (Table 5): coefficients drop and lose significance
- **2002-fixed weights** (Table 6): weaker than cycle-specific across the board
- **Levels** (Table 7): Mayor significant and negative ($\hat{\beta} \approx -0.025$, $F \approx 7.6$)

Main concern: coalition instruments are weak with sector×year FE. Party-level instruments are much stronger. Consider party as primary or investigate why coalition is absorbed by sector×year FE.